

A Cortico-Cerebellar Model for Solving Non-Markovian Environments



Universidad del Desarrollo

Luis Duarte¹, *Principal Investigator*² (ORCID: 0000-0000-0000-0000)

¹Centro de Investigación en Complejidad Social (CICS), Universidad del Desarrollo, Santiago, Chile

²Department, Center, City, Country



CENTRO DE INVESTIGACIÓN EN COMPLEJIDAD SOCIAL

INTRODUCTION

⊢ Cerebellar lesions spare delay conditioning but abolish trace conditioning, revealing a failure to bridge temporal gaps.

⊢ Navigating these non-Markovian gaps requires delayed credit assignment, yet canonical models treat the cerebellum as purely feedforward.

⊢ Feedforward architectures trap the cortex in a Markovian regime restricted to instantaneous observations.

Hypothesis. Passing cortical states through a granular expansion, followed by sparse compression and thalamic return, provides the necessary temporal integration to resolve non-Markovian delays without backpropagation-through-time.

RESULTS

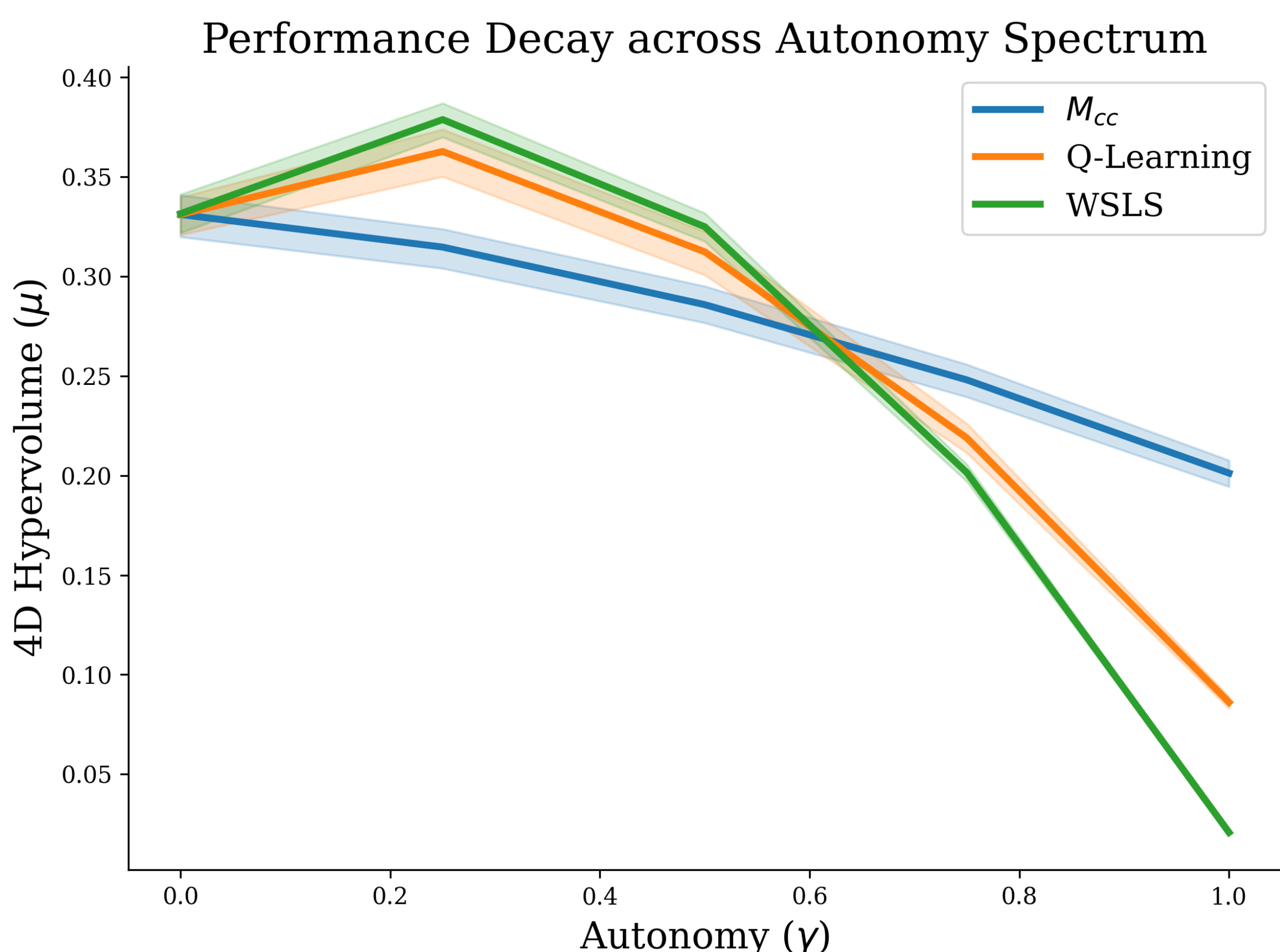


Figure: **NSGA-II Discovery.** Multi-objective evolution jointly minimized distillation and empirical loss to converge on structural bounds.

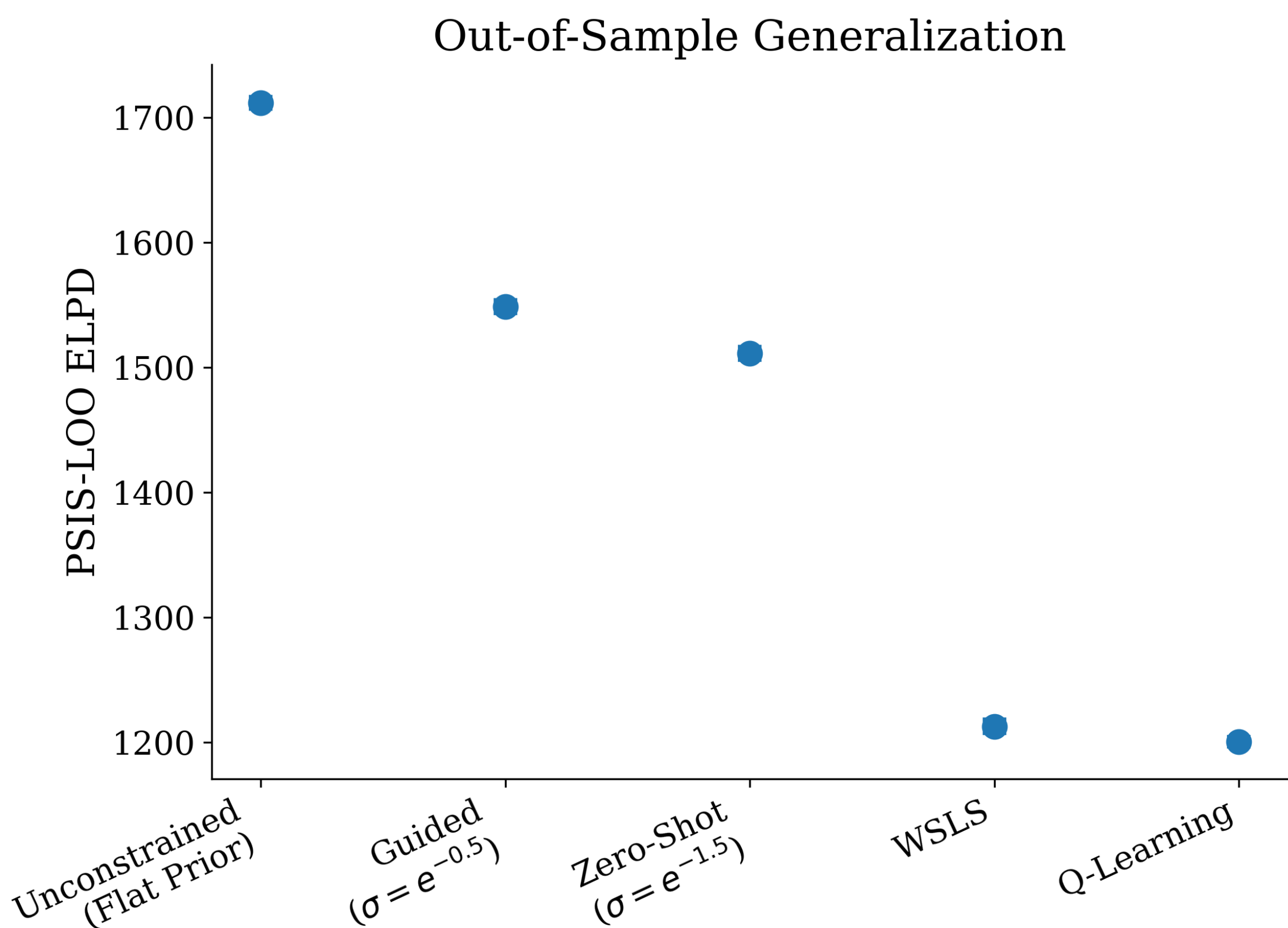


Figure: **Zero-Shot Transfer.** The distilled physiological model predicts raw behavior significantly better than natively optimized heuristics.

Fading Memory Collapse

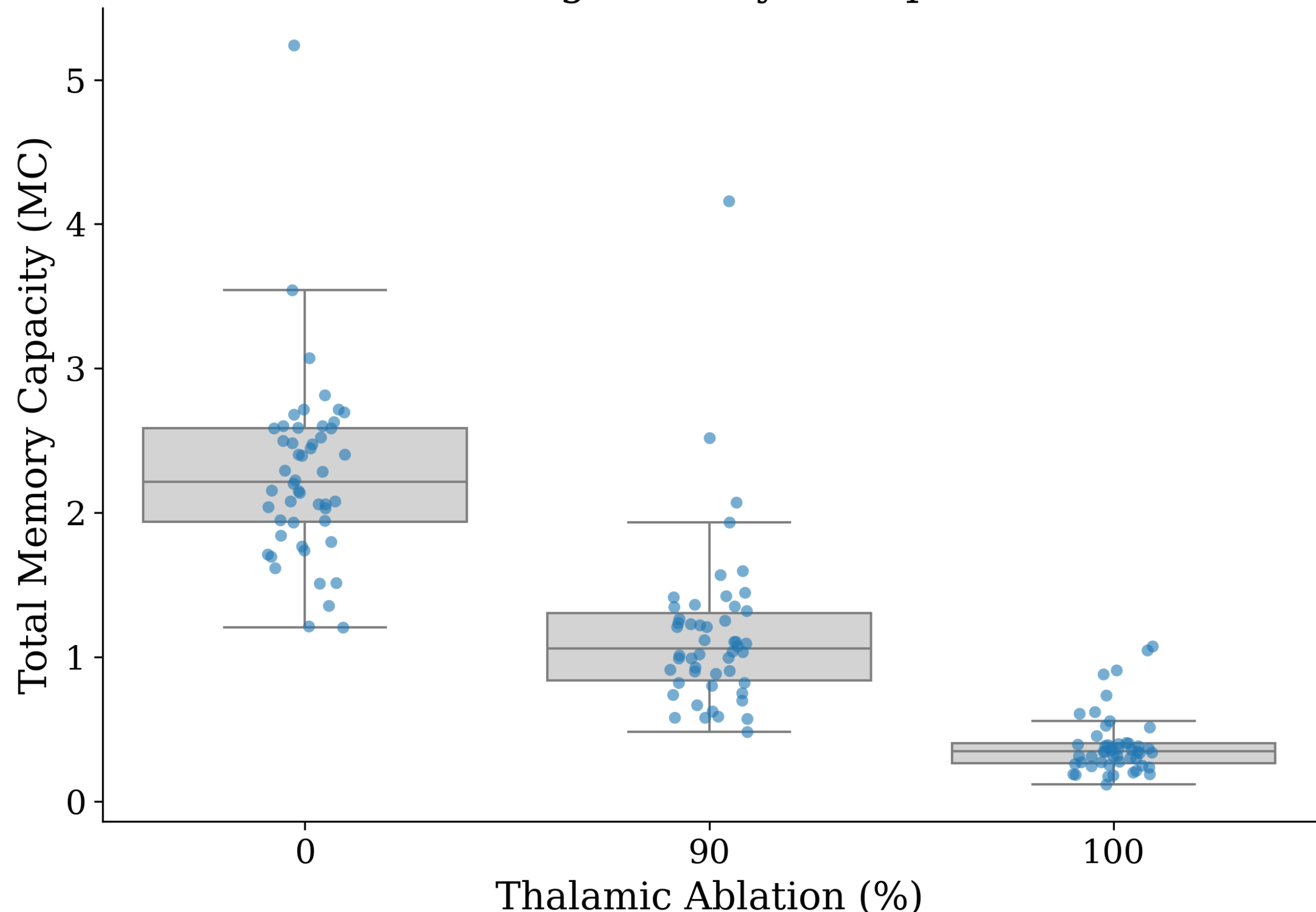


Figure: **Temporal Integration.** The baseline M_{cc} maintains temporal memory, while complete

RESULTS (CONT.)

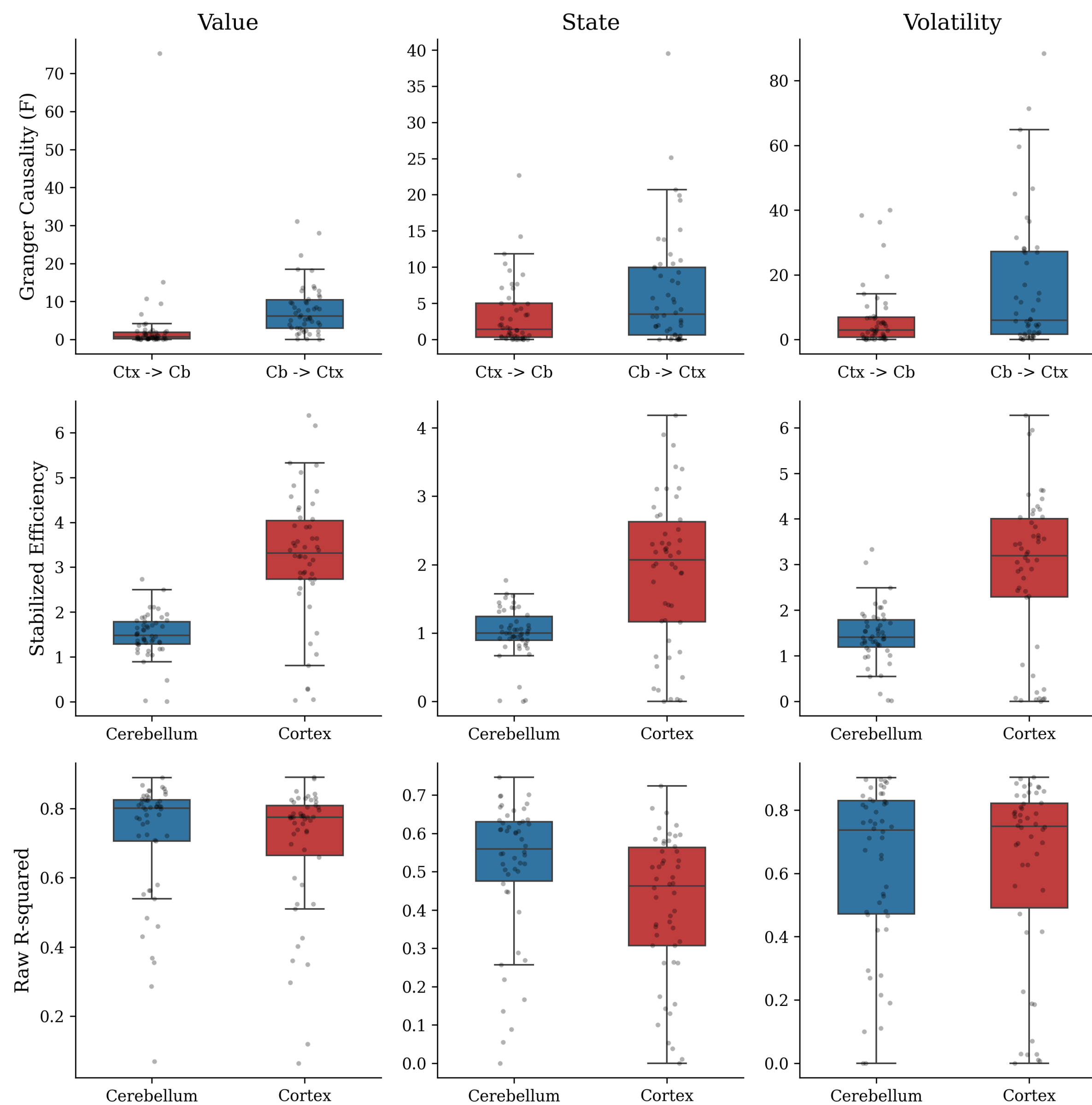


Figure: **Directional Flow.** Granger causality reveals significant causal flow from Cerebellum to Cortex across Value, State, and Volatility.

METHODS

State Expansion & Fading Memory. The granular layer acts as an Echo State Network to generate an eligibility trace.

$$\phi(X_t) = W_{\text{exp}} X_t \quad (1)$$

Where X_t is the cortical state, W_{exp} is the fixed immersion matrix, and $\phi(X_t)$ is the expanded trace.

Sparse Compression. The DCN compresses the trace without geometric collapse.

$$C_t = P_{\theta}(\phi(X_t)) \quad (2)$$

Where P_{θ} is a sparse Achlioptas projection preventing signal cross-talk, and C_t is the compressed historical state returned via the thalamus.

Teacher-Student Distillation. The ODE is evolved via NSGA-II.

$$\mathcal{J}(m) = \mathbb{E}_{\text{Data}}[\log p(\text{RT}, A \mid m)] - \beta \cdot I(\phi(X_t); C_t) \quad (3)$$

Where $\mathcal{J}(m)$ optimizes behavioral fit while penalizing mutual information to enforce sparsity.

CONCLUSION

⊢ Fading memory and cortico-thalamic feedback loops are sufficient to resolve non-Markovian environments.

⊢ Teacher-Student distillation via NSGA-II successfully isolates chaotic physiological ODE parameters without breaking auto-differentiation chains.

⊢ This framework successfully captures emergent value and uncertainty signals purely through historical tracking.

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Scan for References